Link Path Network Optimization Report

Model Type: **link_path**

Metric	Value
Status	Optimal
Objective Value	$250.00
Total Revenue	$250.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$250.00
Accepted Demands	3 / 4
Rejected Demands	1 / 4
Acceptance Rate	75.0%
Nodes	5
Edges	6
Total Initial Capacity	75.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	B -> D (100.0%)
Number of Highly Utilized Edges (≥90%)	1
Most Capacity Expansion	None
Total Lost Revenue	$60.00
Number of Expanded Edges	0

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge B -> D: 100.0% used

Accepted Demands & Flow Paths

Demand 0: A -> E

Bandwidth: 8.0 units **Revenue:** $100.0
Revenue/BW: $12.50/unit

Path	Fraction	Bandwidth
A -> C -> E	100.00%	8.00

Demand 1: A -> D

Bandwidth: 7.0 units **Revenue:** $80.0
Revenue/BW: $11.43/unit

Path	Fraction	Bandwidth
A -> B -> D	57.14%	4.00
A -> C -> D	42.86%	3.00

Demand 2: B -> E

Bandwidth: 6.0 units **Revenue:** $70.0
Revenue/BW: $11.67/unit

Path	Fraction	Bandwidth
B -> D -> E	100.00%	6.00

Rejected Demands

Idx	Source	Target	Bandwidth	Revenue ($)
3	A	E	5.0	60.0

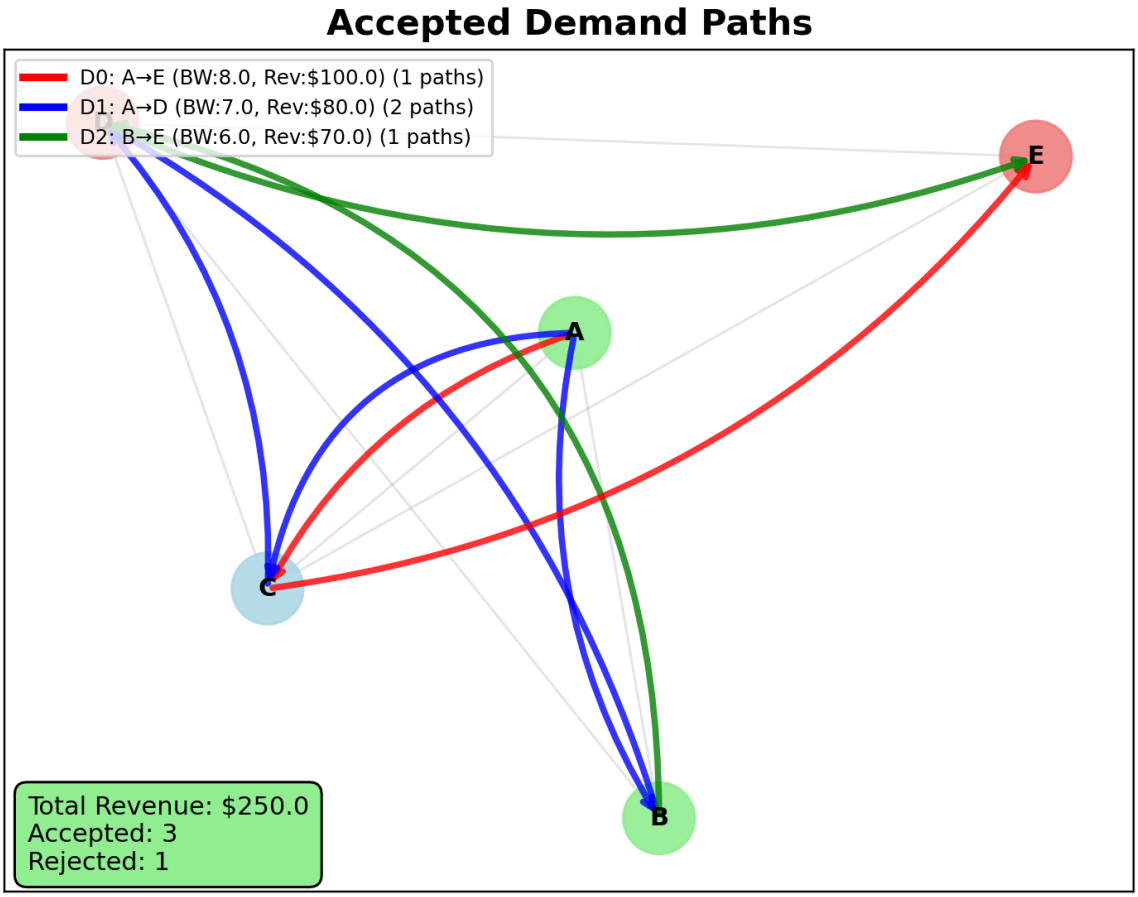
Total Lost Revenue: $60.00

Edge Utilizations (Top 12 Edges)

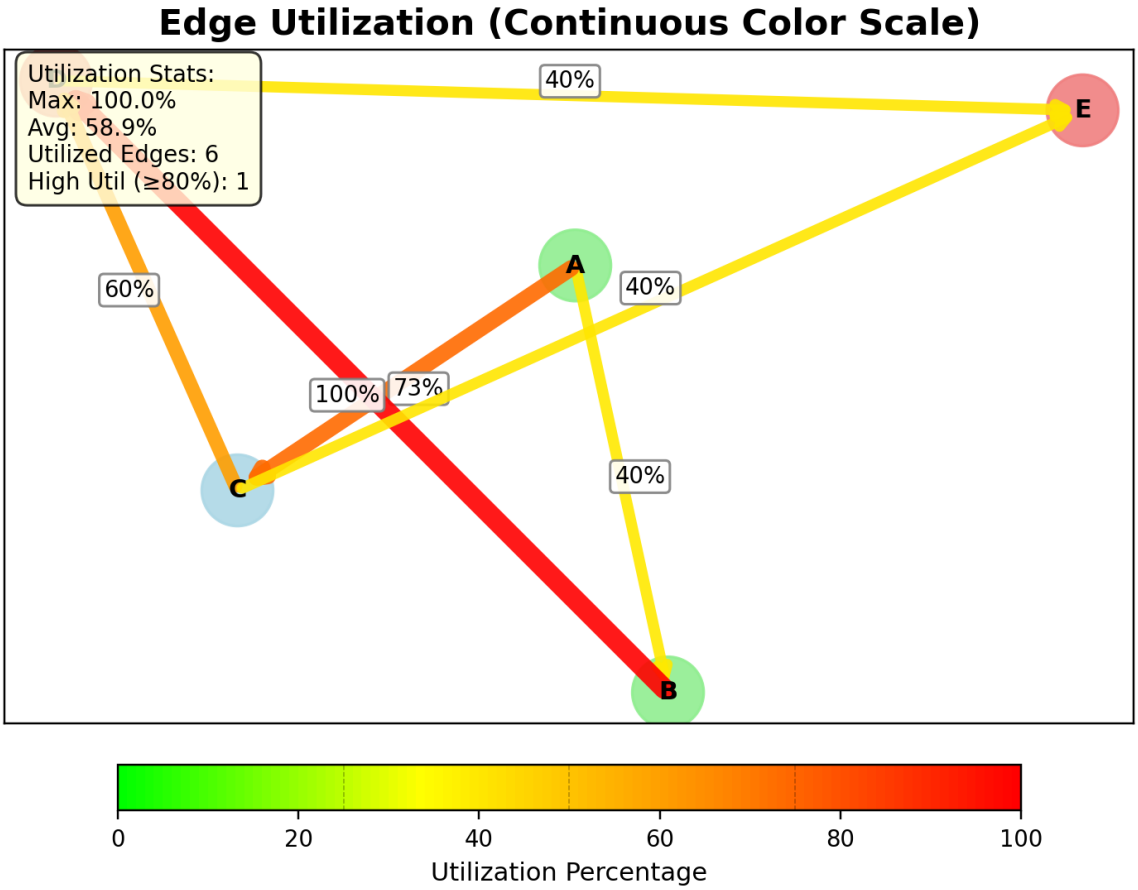
Edge	Flow	Total Capacity	Utilization (%)
B -> D	10.00	10.00	100.0 %
A -> C	11.00	15.00	73.3 %
C -> D	3.00	5.00	60.0 %
C -> E	8.00	20.00	40.0 %
D -> E	6.00	15.00	40.0 %
A -> B	4.00	10.00	40.0 %

Network Analysis Visualizations

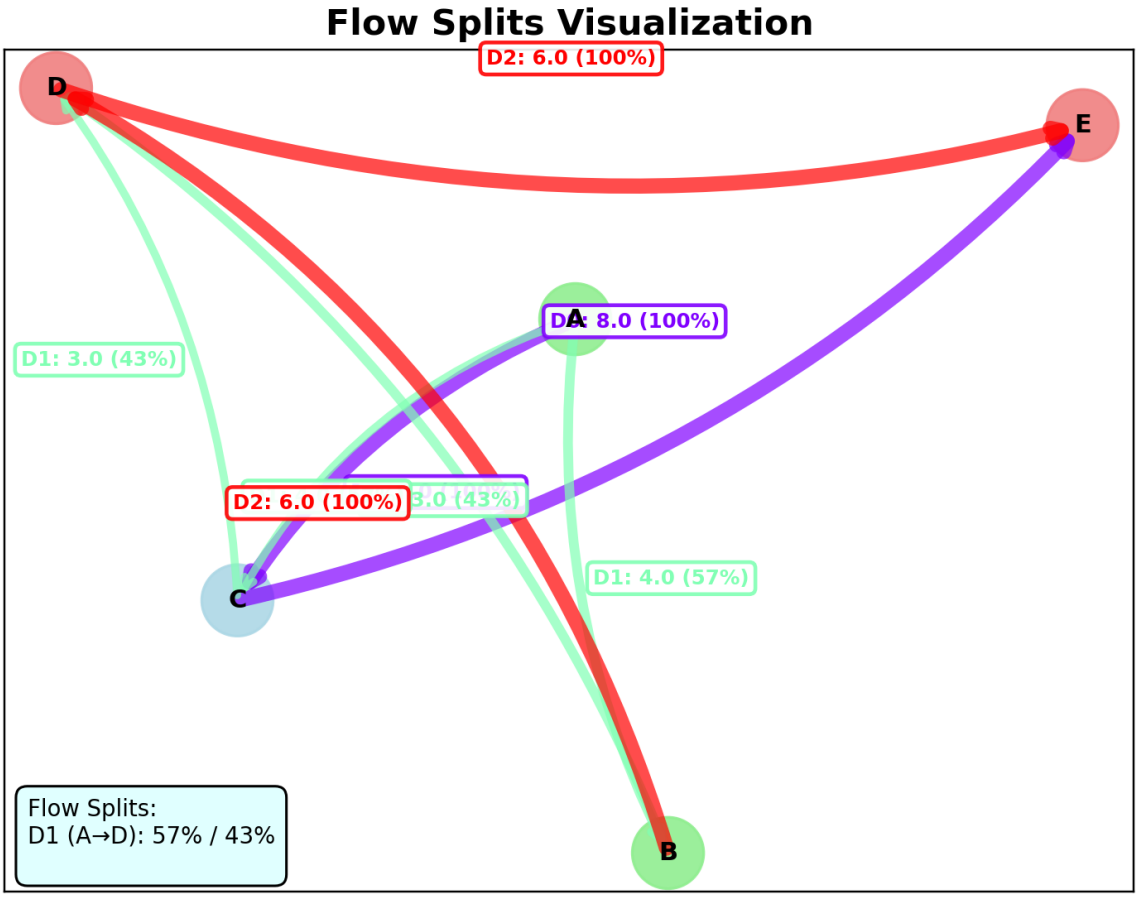
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
D2: 6.0 (100%)
D
E
A
D1: 8.0 (100%)
D1: 3.0 (43%)
D2: 6.0 (100%)
3.0 (43%)
D1: 4.0 (57%)
C
B
Flow Splits:
D1 (A→D): 57% / 43%